

CineStream Content Analytics Dashboard

Module 7 – Final Project Report

Submitted By

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Course: Bachelor of Computer Applications (BCA)

Project Title: CineStream Content Analytics Dashboard

GitHub Repository: https://github.com/sanoobiyaabca25dbi-sys/cinestream_app

Streamlit Dashboard URL: [CineStream Dashboard · Streamlit](#)

1. Introduction

Streaming platforms generate a large amount of data every day. This data contains valuable information about movies, series, documentaries, and stand-up specials. Analysing this data helps companies understand customer preferences, improve recommendations, and make better business decisions.

In this project, an interactive dashboard was developed for a fictional OTT platform called **CineStream** using **Python** and **Streamlit**. The dashboard allows users to filter the catalog and analyse important performance indicators such as views, watch hours, IMDb scores, revenue, and return on investment.

2. Objectives

The main objectives of this project are:

- Clean and prepare the CineStream dataset.
- Build an interactive Streamlit dashboard.
- Display important KPIs and visualisations.
- Provide filtering options for detailed analysis.
- Answer important business questions using dashboard insights.
- Deploy the dashboard online using Streamlit Community Cloud.

3. Technologies Used

- Python
- Streamlit
- Pandas
- NumPy
- Matplotlib
- Plotly

4. Dataset Description

The CineStream dataset contains information about approximately **520 OTT titles** released between **2015 and 2024**.

The dataset includes information such as:

- Content ID
- Title
- Content Type
- Genre
- Language
- Country
- Director
- Release Year
- Added Date
- Runtime
- IMDb Score
- Views
- Watch Hours
- Revenue
- Production Cost
- Subscribers Gained
- Trending Status
- Critic Rating

During preprocessing, the following data cleaning tasks were performed:

- Removed duplicate records.
- Removed trailing spaces.
- Standardised text values.
- Converted AddedDate into datetime format.
- Replaced invalid IMDb and Critic Ratings with missing values.

- Corrected negative subscriber values.
- Created derived columns:
 - Profit_Cr
 - ROI_Pct
 - Performance_Band

5. Dashboard Features

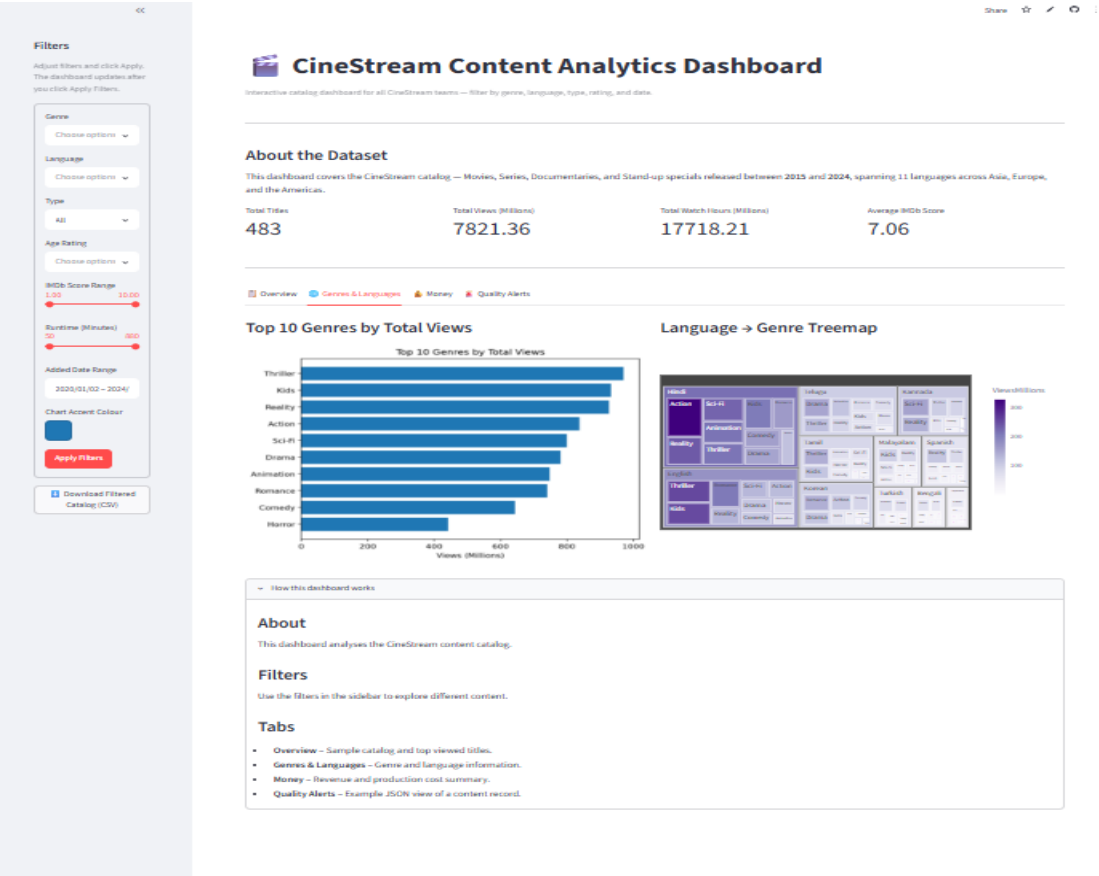
The dashboard includes the following features:

- Interactive sidebar filters
- KPI cards
- Monthly trend analysis
- Content type comparison
- Genre analysis
- Language analysis
- Revenue vs Production Cost analysis
- ROI analysis
- IMDb distribution
- Quality alerts
- Download filtered dataset option
- Responsive charts and status messages

6. Business Questions and Findings

1. Which genres and languages receive the highest views?

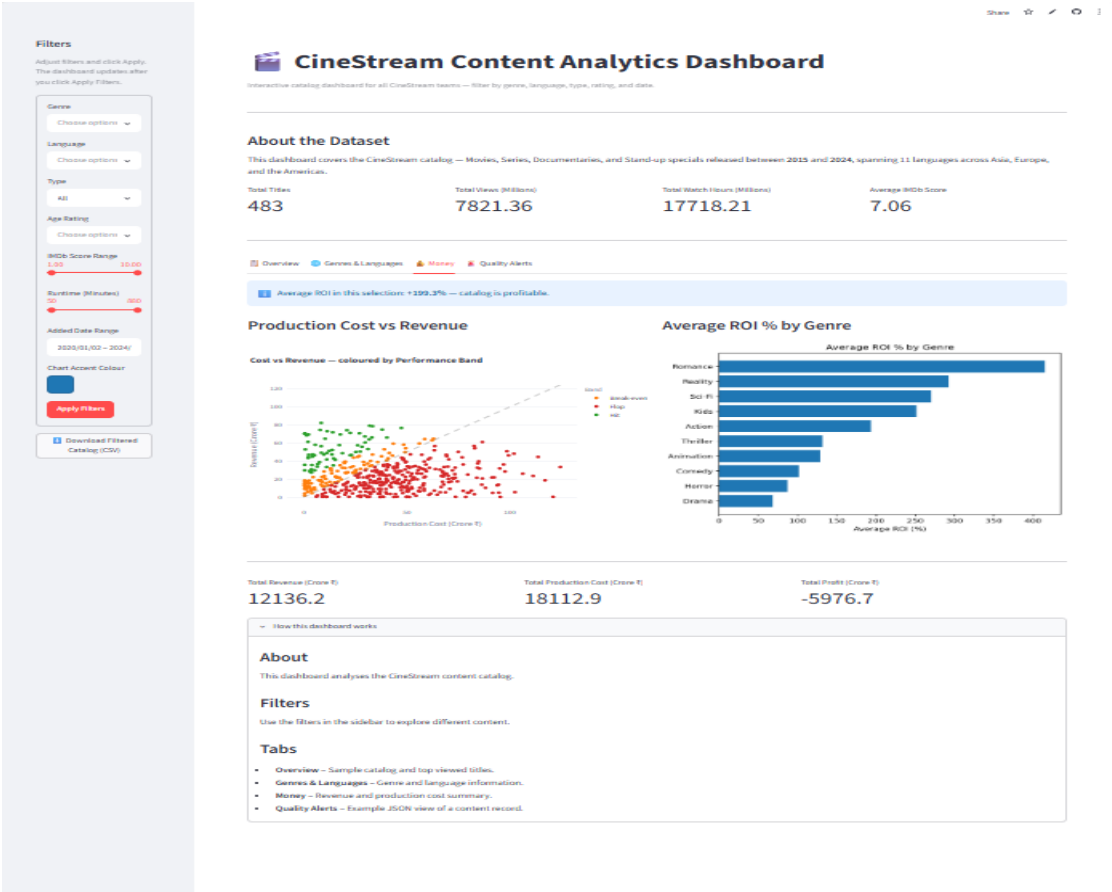
The **Top 10 Genres by Total Views** chart helps identify the genres with the highest audience engagement. The Language Treemap shows how views are distributed across different languages and genres.



2. Are big-budget productions earning back their production cost?

The **Production Cost vs Revenue Scatter Plot** compares production cost with revenue for every title. The ROI chart also shows which genres generate the highest return on investment.

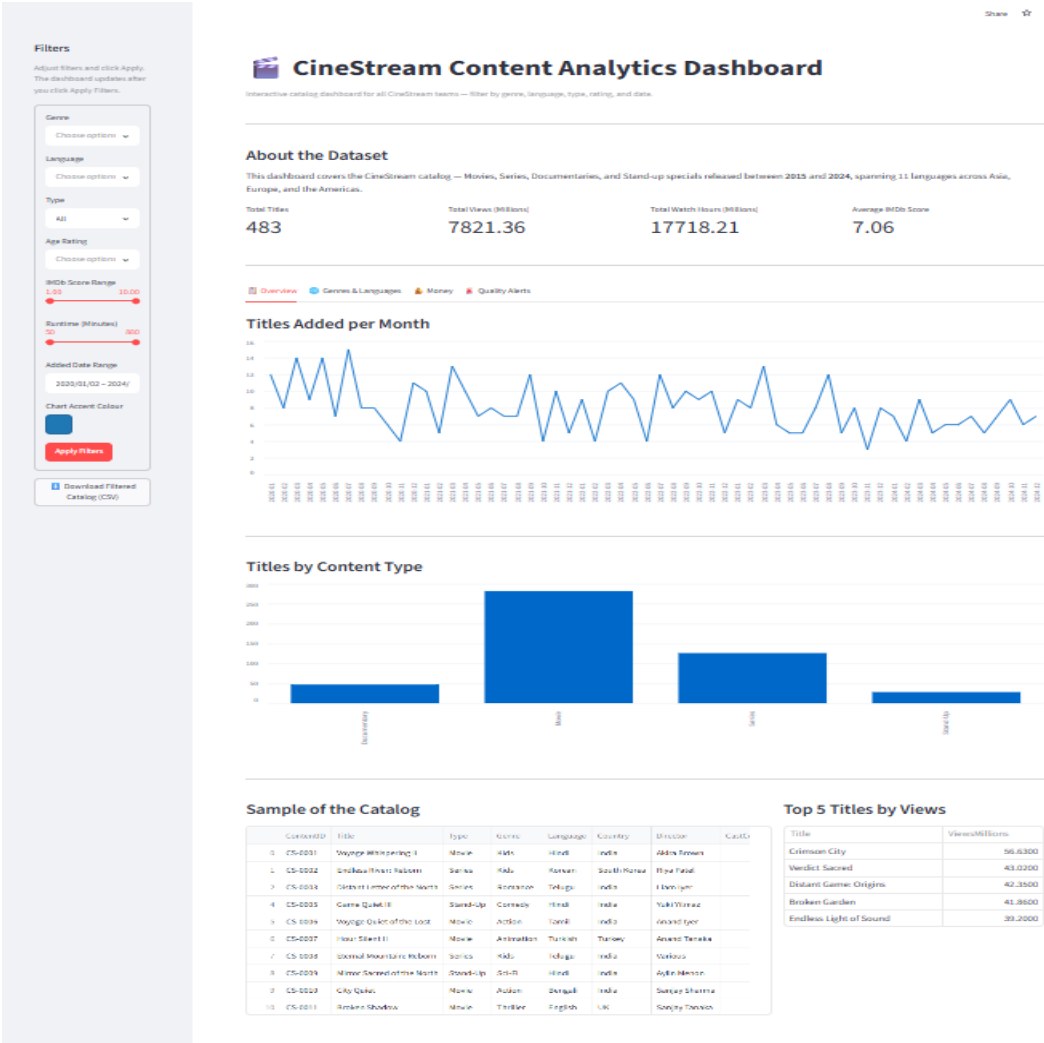
This analysis helps identify profitable and loss-making content.



3. How has the content mix changed over time?

The **Titles Added per Month** line chart and **Content Type Bar Chart** show how different content types such as Movies, Series, Documentaries, and Stand-up specials have been added over time.

This helps understand catalog growth and content trends.



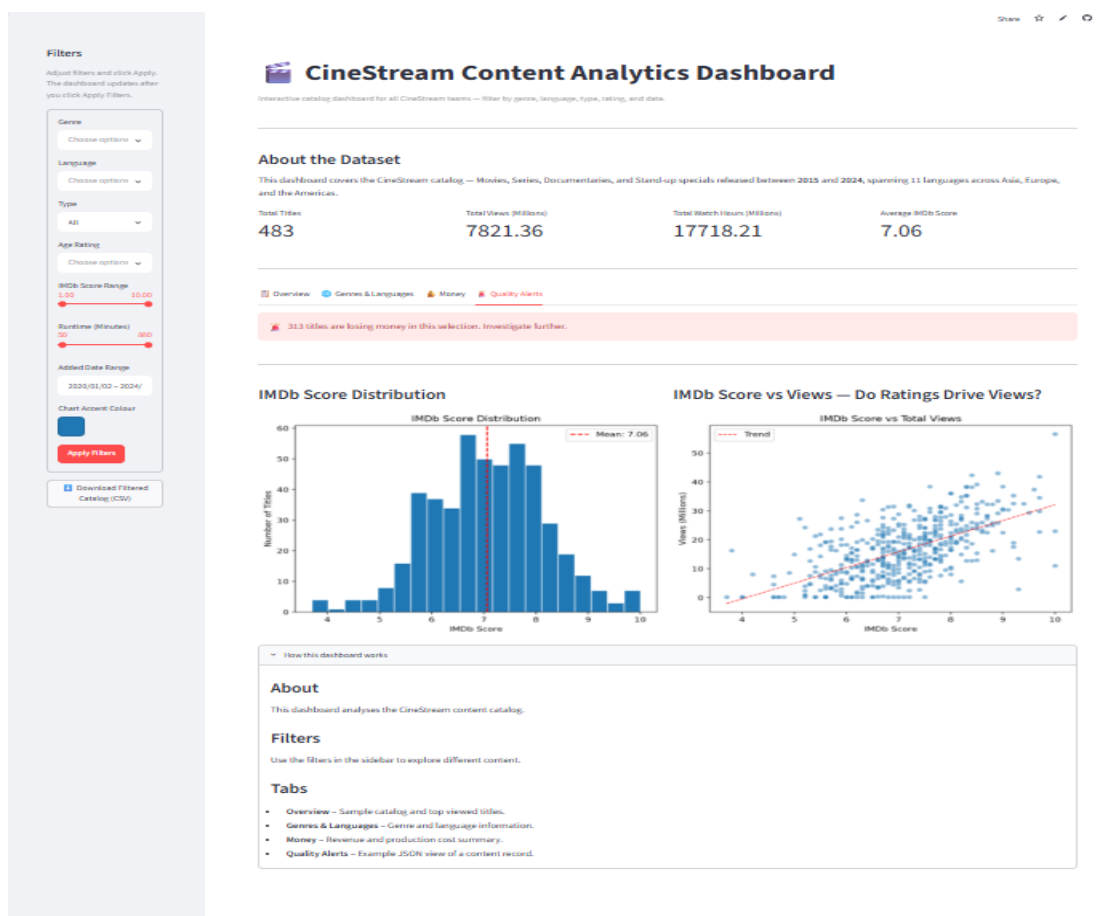
4. Which countries contribute the most subscribers?

The dashboard allows filtering by country and analysing subscriber-related metrics. This helps identify regions that contribute more subscribers and supports future investment decisions.

5. Do higher IMDb scores lead to higher views?

The **IMDb Score vs Views Scatter Plot** compares ratings with audience views.

The trend line indicates whether highly rated titles generally receive more views or whether popularity depends on other factors.



6. Which titles are losing money?

The dashboard calculates **Profit** and **ROI** for every title.

The Quality Alerts section automatically displays warning messages when titles are making losses, helping identify underperforming content quickly.

7. Conclusion

This project demonstrates how data analysis and interactive dashboards can support business decision-making in the OTT industry.

Using Streamlit, Pandas, NumPy, Matplotlib, and Plotly, a complete analytics dashboard was developed that allows users to explore the CineStream catalog through interactive filters and visualisations.

The dashboard successfully answers important business questions related to content performance, audience engagement, profitability, and content trends. It provides an easy-to-use interface that helps decision-makers analyse the catalog without manually inspecting raw data.

Overall, this project improved my understanding of data cleaning, data visualisation, dashboard development, and deployment using Streamlit Community Cloud.

8. References

- [Streamlit Documentation](#)
- [Pandas Documentation](#)
- [NumPy Documentation](#)
- [Matplotlib Documentation](#)
- [Plotly Documentation](#)
- [GitHub](#)